Year 13 Further Maths - Complex numbers

Similar questions - Loci, regions and geometry in the Argand diagram

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Indicate on an Argand diagram, by shading, the region $\{z : |z| \leqslant |z-6|\} \cap \{z : |z-2-\mathrm{i}| \leqslant 3\}$. **[4]**

(b) In an Argand diagram the points representing $1+2\mathrm{i}$ and $3-\mathrm{i}$ are two adjacent vertices of a square S. Find the area of S and all possible pairs of numbers represented by the other two vertices. **[6]**

2

(a) Sketch the locus $C_1 = \{z : 0 \leqslant \arg(z+2\mathrm{i}) \leqslant \frac{1}{3}\pi\}$ on an Argand diagram. **[2]**

(b) Determine whether $1.2-0.8\mathrm{i}$ is in C_1. **[2]**

(c) The locus C_2 is the interior of the circle with centre 3 and radius 2. Use set notation to define C_2 and determine whether $1.2+0.8\mathrm{i}$ lies in it. **[3]**

3 In this question you must show detailed reasoning.

(a) Find the roots of $2z^2-6z+9=0$. **[2]**

The loci C_1 and C_2 are $|z| = |z-3\mathrm{i}|$ and $|z-3| = 3$.

(b) Sketch C_1 and C_2 on one Argand diagram, showing where each meets the imaginary axis. **[3]**

(c) Show that both roots satisfy $|z-3| < 3$. **[2]**

(d) State, with a reason, which root satisfies $|z| < |z-3\mathrm{i}|$, and mark both roots on your diagram. **[3]**

Question 1

(a) $|z| \leqslant |z-6|$ is the half-plane $\mathrm{Re}(z) \leqslant 3$ (points nearer 0 than 6). $|z-2-\mathrm{i}| \leqslant 3$ is the disc with centre $(2,1)$ and radius 3. Shade the part of the disc with $x \leqslant 3$ (the disc and its boundary on the left of the line $x=3$)

(b) The side vector is $(3-\mathrm{i})-(1+2\mathrm{i}) = 2-3\mathrm{i}$, of length $\sqrt{13}$, so the area is 13. Rotating by $\pm 90°$: $\mathrm{i}(2-3\mathrm{i}) = 3+2\mathrm{i}$. Other vertices: $(3-\mathrm{i})+(3+2\mathrm{i}) = 6+\mathrm{i}$ and $(1+2\mathrm{i})+(3+2\mathrm{i}) = 4+4\mathrm{i}$; or $(3-\mathrm{i})-(3+2\mathrm{i}) = -3\mathrm{i}$ and $(1+2\mathrm{i})-(3+2\mathrm{i}) = -2$. Two possible pairs: $\{6+\mathrm{i}, 4+4\mathrm{i}\}$ and $\{-3\mathrm{i}, -2\}$

Question 2

(a) A shaded wedge with vertex at $-2\mathrm{i}$ (the point $(0,-2)$) between the horizontal ray to the right and the ray at $60°$ to the horizontal, both included

(b) $z+2\mathrm{i} = 1.2+1.2\mathrm{i}$ has argument $\frac{\pi}{4} \leqslant \frac{\pi}{3}$ and $\geqslant 0$, so $1.2-0.8\mathrm{i} \in C_1$

(c) $C_2 = \{z : |z-3| < 2\}$. $|1.2+0.8\mathrm{i}-3| = |-1.8+0.8\mathrm{i}| = \sqrt{3.24+0.64} = \sqrt{3.88} < 2$, so it lies in C_2

Question 3

(a) $z = \dfrac{6 \pm \sqrt{36-72}}{4} = \dfrac{3 \pm 3\mathrm{i}}{2}$

(b) C_1 is the horizontal line $\mathrm{Im}(z) = 1.5$ (meets the imaginary axis at 1.5i). C_2 is the circle centre 3, radius 3, which touches the imaginary axis at the origin

(c) $\left|\frac{3\pm 3\mathrm{i}}{2} - 3\right| = \left|\frac{-3\pm 3\mathrm{i}}{2}\right| = \sqrt{\frac{18}{4}} = 2.12 < 3$ for both roots

(d) $|z| < |z-3\mathrm{i}|$ means z is below the line $\mathrm{Im}(z) = 1.5$, so the root $\frac{3}{2} - \frac{3}{2}\mathrm{i}$ satisfies it (the other root, $\frac{3}{2} + \frac{3}{2}\mathrm{i}$, has $\mathrm{Im} = 1.5$, which is on the line, so it does not satisfy the strict inequality)